

Source: Local Subject Media

24CS206 WEB TECHNOLOGIES

Hours Per Week :	L	T	P	SL	C
	0	2	2	0	2

Prerequisite: Object-Oriented Programming (Java or Python)

Course Description:

This course bridges the gap between traditional full-stack web development and modern artificial intelligence. It offers comprehensive training in building dynamic web applications using HTML5, CSS3, modern JavaScript (ES6+), and Node.js, while seamlessly integrating AI capabilities. Students will learn to leverage AI coding assistants, implement prompt engineering, integrate Large Language Model (LLM) APIs, and utilize vector databases. The ultimate objective is to empower students to design, develop, and deploy intelligent, agentic web applications that can understand, process, and generate dynamic content.

MODULE-1: MODERN WEB FOUNDATIONS & CLIENT-SIDE INTELLIGENCE

Flow Logic: We start by building static pages, add logic with JS, introduce modern frontend frameworks, and then inject AI directly into the user's browser.

UNIT-1 **0L+6T+6P+0SL=12 Hours**

CORE WEB TECHNOLOGIES & AI-ASSISTED DEVELOPMENT

Modern HTML5 & CSS3: Semantic HTML, Canvas, Audio and Video Integration, CSS3 Grid, Flexbox, Responsive Design, CSS Variables. JavaScript (ES6+): DOM Manipulation, Arrow Functions, Promises, Async/Await, Closures, Fetch API, Event Handling, Regular Expression Validation. Code Debugging, Boilerplate Code Generation. [TB2: Chapters 1-9 & 10-15], [RB1: Chapters 3 & 11], [WL1: JavaScript Guide], [WL3: AI Assistant Setup Docs]

UNIT-2 **0L+6T+6P+0SL=12 Hours**

CLIENT-SIDE FRAMEWORKS & BROWSER AI

Frontend Frameworks: Component Architecture, State Management, Props and Data Binding, Routing (Angular/React). Intelligent UI/UX: Conversational Interfaces, Chat UI, Adaptive Dynamic Forms. Browser-Based AI: Web Speech API, Speech-to-Text, Text-to-Speech, Browser-based Pre-trained Models, TensorFlow.js, Image Classification, Client-side Sentiment Analysis. [TB1: Chapters 17-20], [TB3: Chapters 2, 4 & 8], [RB2: Chapter 4], [WL2: Component Architecture Docs]

PRACTICE:

Design a responsive, multi-page personal portfolio using HTML5 and CSS3. Utilize an AI coding assistant to generate the initial scaffolding, and manually refine the code for optimal layout.

Develop an interactive catalog page for books or products using JavaScript. Implement client-side filtering and sorting functionalities that update the DOM without requiring page reloads.

Design a comprehensive student registration form. Validate the inputs using standard JavaScript regular expressions and integrate a lightweight client-side AI model to detect inappropriate language in the text areas.

Create a voice-controlled web page featuring a color palette. Utilize the Web Speech API so that when a user speaks a command (e.g., "Make background blue"), the web page background updates accordingly.

Develop a web application that captures an image via the user's webcam and utilizes pre-trained TensorFlow.js models in the browser to identify and label the primary object in the frame.

MODULE-2 SERVER-SIDE DATA & INTELLIGENT WEB AGENTS

Flow Logic: We build a secure backend server first, then connect that server to LLMs and Vector DBs to give it "brains," and finally upgrade the application to act as an autonomous agent before deploying.

UNIT-1 **0L+6T+6P+0SL=12 Hours**

SECURE BACKEND DEVELOPMENT & DATABASES

Node.js & Express: Node.js Server Setup, Express Routing, Middleware, RESTful APIs, CORS. Authentication & Sessions: JSON Web Tokens (JWT), Secure Login/Signup Flows, Protected Routes, Session Management. Database Integration: NoSQL Fundamentals, MongoDB, CRUD Operations, Mongoose ODM. [TB1: Chapters 5, 6, 8, 13, 14 & 26], [WL1: HTTP Authentication Guide]

UNIT-2 **0L+6T+6P+0SL=12 Hours**

LLM INTEGRATION & CONTEXTUAL MEMORY

API Integration: OpenAI API, Google Gemini API, Secure API Key Management, Environment Variables. Prompt Engineering: System Prompts, Few-shot Prompting, Context Window Management, AI Hallucination Mitigation. Vector Databases & RAG: Data Embeddings, Vector Database Setup (Pinecone/Weaviate/MongoDB Atlas Vector Search), Semantic Data Storage. [RB2: Chapters 2 & 5], [WL3: Node.js Quickstart], [WL4: Q&A; with RAG], [WL5: Understanding Embeddings]

UNIT-3 **0L+6T+6P+0SL=12 Hours**

BUILDING AGENTIC TOOLS & DEPLOYMENT

Agentic Web Workflows: Autonomous AI Tasks, Tool Calling, Automated Web Scraping, Content Summarization Pipelines. Deployment & Security: Application Hosting (Vercel/AWS EC2), Secure Environment Configurations, Prompt Injection Security, Endpoint Rate Limiting. [TB1: Chapter 32], [WL4: Agents and Tools], [WL2: Deployment Guides]

PRACTICE:

Design a RESTful API utilizing a Node.js and Express backend connected to a MongoDB database to perform complete CRUD operations for a library management system.

Implement a secure authentication system featuring login and signup flows using JSON Web Tokens (JWT) to protect specific application routes and manage user sessions.

Develop a custom frontend chatbot interface connected to a Node.js backend. Securely format user inputs, make calls to an LLM API, and stream the generated AI responses back to the client interface.

Design a semantic search engine by embedding and storing a dataset of university course descriptions in a vector database. Allow users to search for courses using natural language queries rather than exact keyword matches.

Develop an intelligent web agent (Capstone Application) where a user can upload a document or provide a URL. Program the backend to extract the text, utilize an LLM to generate a summarized report, translate it into a specified language, and render the final results on a secure user dashboard.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
CO1	Understand the foundational concepts of modern web technologies, including semantic HTML5, CSS3, JavaScript (ES6+), and AI-assisted development tools.	Understand	M-I U-I	PO1, PO2, PO5
CO2	Describe the component architecture of modern front-end frameworks and the mechanisms of browser-based AI APIs like Web Speech and TensorFlow.js.	Understand	M-I U-II	PO1, PO2
CO3	Apply Node.js, Express, and NoSQL databases to build and secure server-side RESTful APIs with JWT authentication.	Apply	M-II U-I	PO1, PO2, PO3, PO5

SKILLS:

- ✓ *Develop functional web and mobile applications using modern AI coding assistants.*
- ✓ *Design effective prompt engineering workflows for architecture and UI generation.*
- ✓ *Integrate frontend components with backend databases, APIs, and cloud services.*
- ✓ *Debug, test, and safely deploy AI-generated application code.*

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
CO4	Analyze prompt engineering strategies and Retrieval-Augmented Generation (RAG) architectures for integrating Large Language Models and vector databases.	Analyze	M-II U-II	PO1, PO2, PO3, PO4
CO5	Design and deploy comprehensive, multi-tier intelligent web applications that execute autonomous agentic workflows while addressing endpoint security.	Create	M-II U-III	PO2, PO3, PO5, PO8

TEXTBOOKS (T):

Brad Dayley, Node.js, MongoDB and Angular Web Development, 2nd Edition, Addison-Wesley Professional, 2017.

Jon Duckett, HTML and CSS: Design and Build Websites (Part 1) & JavaScript and JQuery: Interactive Front-End Web Development (Part 2), Wiley.

Gant Laborde, Learning TensorFlow.js: Powerful Machine Learning in JavaScript, O'Reilly Media, 2021.

REFERENCES (R):

Cay S. Horstmann, Modern JavaScript for the Impatient, Addison-Wesley Professional, 2020.

Olivier Caelen, Marie-Alice Blete, Developing Apps with GPT-4 and ChatGPT: Build Intelligent Chatbots, Content Generators, and More, O'Reilly Media, 2023.

Web Links & Official Documentation (W):

MDN Web Docs (Mozilla Developer Network) - Standard for HTML, CSS, JS, and Web APIs.

(https://developer.mozilla.org)

React Official Documentation. (https://react.dev)

Google Gemini API Documentation / OpenAI API Documentation. (https://ai.google.dev/docs | https://platform.openai.com/docs)

LangChain.js Documentation - Standard for RAG and Agentic workflows. (https://js.langchain.com/docs)

Pinecone / MongoDB Atlas Vector Search Docs. (https://docs.pinecone.io | https://www.mongodb.com/docs/atlas/atlas-vector-search)